Autonomous Aerospace Monitoring Assistant:

An Integrative AI-Based Anomaly Triage Framework

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Integrative Industry Synthesis Capstone

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Location Of The Integrated Industry Artifact

The integrated industry artifact required for Project 7 is the fully executed Jupyter notebook contained in the accompanying repository.

GitHub Repository:

https://github.com/MinervaRose/industry-integrated-ai-systems-synthesis

The notebook implements Tasks 1–5 of the project instructions, including:

- Industry selection and problem definition
- Integration plan across prior capstone projects
- Integrated AI system architecture and design
- Applied artifact implementation (end-to-end pipeline)
- System evaluation and scenario-based analysis

All supporting materials referenced in this paper, including the architecture diagram, `requirements.txt`, and presentation slides, are included in the same repository.

Overview And Industry Context

Modern aerospace operations depend on continuous streams of telemetry and visual sensor data to monitor the health, performance, and safety of satellites and spacecraft. These systems operate in environments where failure is costly, delays are dangerous, and decisions must often be made under uncertainty. Anomalies—unexpected deviations in telemetry signals or sensor imagery—can indicate benign environmental effects, instrumentation noise, or critical system malfunctions. The challenge lies not only in detecting anomalies but in triaging them responsibly, efficiently, and transparently.

Traditional rule-based systems struggle with high-dimensional sensor data and complex nonlinear interactions. Machine learning and deep learning methods, by contrast, are well suited to modeling structured telemetry data and image-based signals (Bishop, 2006; Goodfellow et al., 2016). However, in safety-critical industries such as aerospace, pure automation is insufficient. Decisions must be auditable, interpretable, and accountable. Overreliance on automated systems can introduce automation bias, obscure responsibility, and increase systemic risk (Varshney, 2016).

The system designed in this capstone addresses this problem space by proposing an integrative AI-based anomaly triage framework. Rather than focusing on a single predictive model, the system integrates multiple AI components—structured data classification, deep learning-based image analysis, generative modeling, and agentic orchestration—into a cohesive decision-support pipeline. The objective is not to replace human operators but to augment them with structured, transparent, and ethically bounded decision support.

This problem is appropriate for AI-based solutions because it involves high-volume multimodal data, probabilistic reasoning under uncertainty, and the need for adaptive, data-driven pattern recognition. At the same time, it demands careful consideration of safety, transparency, and human oversight.

Integration Rationale

A central objective of this capstone was to demonstrate system-level integration across prior coursework rather than to build another isolated model. The final system integrates four core domains:

1. Structured machine learning for telemetry anomaly detection
2. Deep learning for image-based signal analysis
3. Generative modeling using a variational autoencoder (VAE)
4. Agentic decision logic for triage orchestration

Each component draws from prior capstone work and theoretical foundations.

The structured telemetry classifier represents prior work in supervised machine learning, grounded in probabilistic pattern recognition principles (Bishop, 2006). Such models are well suited to tabular telemetry features and produce calibrated confidence scores that inform downstream routing.

The image classifier component builds on deep convolutional architectures for image recognition (Goodfellow et al., 2016). In aerospace contexts, visual inspection of satellite imagery or sensor outputs remains critical. Deep learning provides robust feature extraction and classification for high-dimensional image inputs.

The generative probe component draws inspiration from variational autoencoders (Kingma & Welling, 2014). VAEs learn latent representations of normal operating conditions and can simulate perturbations to assess robustness. In this system, the generative probe is triggered under uncertainty, enabling stress-testing of borderline anomaly cases.

Finally, the triage agent orchestrates decision-making. This agentic layer aggregates model outputs, applies rule-based thresholds, generates explanations, and determines escalation pathways. This aligns with principles of human-AI interaction, emphasizing transparency, appropriate automation boundaries, and support rather than replacement of human expertise (Amershi et al., 2019).

The rationale for combining these components is that aerospace anomaly triage is inherently multimodal and risk-sensitive. No single model is sufficient. Structured telemetry data and image signals provide complementary evidence. Generative modeling offers robustness testing. Agentic logic ensures that the system's behavior aligns with operational and ethical constraints.

The integration thus reflects a layered defense strategy: detection, corroboration, stress-testing, and human-informed escalation.

System Design And Technical Decisions

The architecture consists of five primary modules:

1. Data Intake and Validation
2. Structured Telemetry Classifier
3. Image Classifier
4. Generative Robustness Probe
5. Triage Agent and Audit Logger

Data flows from sensor inputs into parallel predictive branches: structured telemetry and image analysis. Each branch outputs a classification label and confidence score. These outputs are aggregated into a unified evidence score. If confidence falls below defined thresholds, the generative probe is triggered to simulate perturbations and assess robustness.

The triage agent then applies decision rules based on aggregated confidence, anomaly severity, and data quality flags. Possible outcomes include:

- Clear and proceed
- Monitor
- Escalate to human review

A key technical decision was to design the classifiers as modular components representing pretrained artifacts from prior capstone work. The focus of this artifact is integration and orchestration rather than retraining. This choice aligns with the capstone's emphasis on applied system design rather than isolated model development.

Confidence thresholds were selected to illustrate tradeoffs between false positives and false negatives. In safety-critical contexts, false negatives (missed anomalies) are typically more costly than false positives. However, excessive escalation can overwhelm operators and reduce trust. This tension reflects well-known safety engineering tradeoffs in machine learning systems (Varshney, 2016).

An audit logging mechanism records each decision pathway, including confidence scores, routing choices, and final actions. This supports traceability and accountability. System boundaries are explicitly defined: the AI does not autonomously execute mission-critical commands; it provides structured recommendations.

This design reflects principles of layered reliability and bounded automation.

Ethical Considerations And Responsible Ai

Safety-critical AI systems must balance performance with ethical responsibility. Several ethical concerns arise in this domain.

Automation Bias

Human operators may over-trust AI recommendations, especially when outputs appear confident or authoritative. Amershi et al. (2019) emphasize the importance of designing systems that appropriately calibrate trust. In this architecture, uncertainty triggers escalation rather than silent automation. Confidence scores and explanatory outputs are surfaced to the user, reducing blind reliance.

Accountability And Transparency

In aerospace contexts, accountability is paramount. The audit logging system ensures traceability of decision pathways. Transparency aligns with ethical frameworks that emphasize explicability as a core AI principle (Floridi et al., 2018). By preserving intermediate outputs and routing decisions, the system supports post hoc analysis and governance.

Bias And Model Limitations

Machine learning models may inherit biases from training data. Although this prototype does not retrain models, the design explicitly acknowledges dataset representativeness as a constraint. Responsible deployment would require validation across diverse operating conditions and mission scenarios.

Safety And Risk Management

Varshney (2016) argues that engineering safety into machine learning requires anticipating failure modes. This system incorporates multiple safeguards: multimodal corroboration, uncertainty routing, generative stress testing, and mandatory human review for high-risk cases.

Governance And Deployment Boundaries

The system is explicitly positioned as decision support, not autonomous control. This boundary reduces moral hazard and clarifies responsibility. Floridi et al. (2018) emphasize that AI systems should enhance human flourishing and respect human agency. By preserving human-in-the-loop escalation, the design aligns with these principles.

Overall, ethical considerations are not appended after technical design; they shape architecture choices directly.

Lifecycle Governance And Continuous Monitoring

Responsible AI deployment extends beyond initial design. In aerospace contexts, system performance must be continuously monitored and recalibrated as operational conditions evolve. Dataset drift, sensor degradation, and environmental shifts can alter model reliability over time. As Goodfellow et al. (2016) note, deep learning systems can generalize poorly outside their training distribution.

Accordingly, a responsible deployment framework would require periodic retraining validation, human audit checkpoints, and version-controlled logging of model updates. Governance policies should define escalation thresholds, override authority, and accountability chains to prevent ambiguity in responsibility allocation (Floridi et al., 2018).

Embedding governance within the lifecycle ensures that ethical reasoning remains operational rather than theoretical.

Evaluation And Reflection

Evaluation focused on system-level behavior across simulated scenarios. The objective was not to maximize model accuracy but to assess integration logic, routing robustness, and decision coherence.

What Worked

The modular architecture proved flexible and interpretable. The triage agent provided structured, explainable outputs. Conditional triggering of the generative probe allowed dynamic escalation under uncertainty. The audit logging mechanism enhanced traceability and improved clarity during scenario analysis.

Integration across domains strengthened robustness. Multimodal evidence aggregation reduced reliance on single-model predictions, illustrating the value of cross-domain design.

Limitations

Because the artifact emphasizes integration, the classifiers are represented as pretrained modules rather than retrained within this notebook. Real-world deployment would require rigorous validation, calibration, and dataset governance.

Generative robustness testing is conceptually powerful but computationally expensive in production environments. Latency and compute constraints would need careful consideration.

Finally, threshold tuning remains a delicate tradeoff. Overly conservative thresholds increase operational burden; overly permissive thresholds increase risk.

Failure Modes And Risk Scenarios

In safety-critical AI systems, evaluation must extend beyond performance metrics to structured failure analysis. Several potential failure modes were considered in the design of this anomaly triage system.

First, correlated model error presents a systemic risk. If both the structured telemetry classifier and the image classifier are trained on overlapping operational assumptions, they may fail simultaneously under novel environmental conditions. This highlights the importance of dataset diversity and cross-validation under mission-specific edge cases.

Second, overconfident misclassification represents a critical hazard. High-confidence false negatives could suppress escalation pathways, delaying human review. To mitigate this risk, the architecture incorporates uncertainty thresholds and optional generative robustness probing, ensuring that borderline predictions are stress-tested before final routing.

Third, generative probe misuse could introduce computational inefficiency or misinterpretation. While variational autoencoders provide valuable latent space analysis (Kingma & Welling, 2014), their outputs must not be treated as ground truth simulations. In production deployment, probe outputs would require validation against domain physics constraints.

This analysis reinforces that integration complexity introduces new system-level risks that must be engineered deliberately rather than assumed away.

Lessons Learned

The most significant insight from this capstone was that integration complexity quickly exceeds model complexity. Designing orchestration logic, defining boundaries, and anticipating failure modes require more strategic reasoning than training a standalone model.

Cross-domain fluency is essential. Machine learning, deep learning, generative modeling, and agentic reasoning each contribute differently. True system-level understanding lies in how they interact.

Ethical reasoning cannot be retrofitted. It must inform architecture decisions from the outset.

Professional Relevance

This project demonstrates readiness for real-world AI roles by emphasizing system-level thinking, responsible deployment, and integration across domains. In professional contexts, AI practitioners rarely build isolated models; they design pipelines, coordinate stakeholders, and manage risk.

The artifact showcases:

- Architectural design skills
- Multimodal integration
- Human-in-the-loop reasoning
- Ethical foresight

- Evaluation under uncertainty

These competencies align with applied AI engineering, AI consulting, and AI governance roles. By explicitly addressing tradeoffs, constraints, and safeguards, the project reflects professional judgment rather than purely academic experimentation.

Applied Skill Translation

This project demonstrates competencies directly aligned with AI systems engineering and responsible AI consulting roles. The architecture design reflects the ability to decompose complex problems into modular components, define interaction contracts between models, and establish auditability mechanisms. The explicit tradeoff analysis mirrors decision-making processes required in safety-critical industries such as aerospace, healthcare, and autonomous systems.

Furthermore, the integration of technical modeling with governance considerations reflects interdisciplinary readiness. Modern AI roles increasingly demand collaboration across engineering, policy, and operations teams. By embedding ethical safeguards, human-in-the-loop boundaries, and structured failure analysis, this work demonstrates the capacity to operate not merely as a model builder but as a system designer responsible for risk management and stakeholder impact.

References

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